



A Study of the Small Inductive Dimension in the Area of Finite Lattices

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Abstract

The Dimension Theory in the class of frames was the base of many researches. Especially, the covering dimension, the quasi covering dimension, the small inductive dimension and the large inductive dimension for frames have been studied extensively (see (Banaschewski and Gilmour J. London Math. Soc. **39**(2) 1–8 [1989](#)), (Brijlall and Baboolal Indian J. Pure Appl. Math. **39**(5), 375–402 [2008](#)), (Brijlall and Baboolal Indian J. Pure Appl. Math. **41**(3), 535–553 [2010](#)), (Charalambous J. London Math. Soc. **8**(2), 149–160 [1974](#)), (Charalambous Topology Proc. **20**, 49–66 [1995](#)), (Georgiou et al. Topology Appl. **275**, 106995–13 [2020](#))). Moreover, the study of the meaning of dimension in Order and Lattice Theory tended to the study of various dimensions, such as the order dimension for finite posets and the covering dimension for finite lattices (see, for example, (Boyadzhiev et al. Appl. Math. Comput. **333**, 276–285 [2018](#)), (Dube et al. Discrete Math. **338**(7), 1096–1110 [2015](#)), (Georgiou et al. Quaest. Math. **39**(6), 797–814 [2016](#))). In this paper, we study the notion of the small inductive dimension, ind , in the class of finite lattices. We provide a new characterization of this dimension, using the so-called minimal covers, and investigate properties of this dimension. We study the small inductive dimension of the linear sum, the Cartesian product, the lexicographic product and the rectangular product of two finite lattices and the sublattice property. Also, we present various examples, investigating the relation of the small inductive dimension with the Krull and covering dimension.

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1 Introduction

The Dimension Theory is a branch of Topology that has been studied extensively, defining some of the most known dimensions, the covering dimension, the small inductive dimension and the large inductive dimension (see, for example, [10–14]). Moreover, these dimensions have been studied in the realm of finite and Alexandroff spaces (see, for example, [15–18]).

However, many researchers such as M. Charalambous, G. Gilmour, B. Banaschewski, D. Brijlall, D. Baboolal made a significant effort to develop a Frame Dimension Theory, defining the covering dimension, the small inductive dimension and the large inductive dimension in classes of frames (see [1–5]).

Simultaneously, various dimensions have been studied in the realm of finite partially ordered sets and lattices. The order dimension, the Krull dimension and the covering dimension are examples of such dimensions (see [7–9, 19, 20]).

The term finite set is one of the main keys of Discrete Mathematics. Finite sets have many important applications to computer science, algorithms, programming languages, cryptography and software development. This fact tends us to pay attention to the meaning of finite. At the same time, since Digital Topology studies the topological properties of digital images and discrete arrays in two or more dimensions, the notions of dimensions, and more specifically topological dimensions, win their own important role. Moreover, partially ordered sets and especially lattices are used to investigate more properties of digital images (see for example [21]).

On the other hand, from a classical topological point of view, topological spaces like spectral spaces have been studied under the prism of lattices [22]. This fact led to insert definitions of topological notions in terms of lattices and more specifically finite lattices. The meaning of dimension is one of these topological notions. As we have mentioned, the order dimension, the Krull dimension and the covering dimension have been inserted and studied in Lattice Theory. Therefore, a research issue that arises is the study of dimensions for lattices. The combination of finite lattices and their dimensions provides new results which develop the current research.

In this paper, we study the notion of the small inductive dimension, ind , in the class of finite lattices, proving characterizations and properties of this dimension. Especially, in Section 2, we give a description of the small inductive dimension in details, giving a new characterization of this dimension, using the so-called minimal covers and studying the sublattice property. In Section 3, we study the small inductive dimension of the linear sum, the Cartesian product, the lexicographic product and the rectangular product of two finite lattices and in Section 4, we present various examples, investigating the difference between the small inductive dimension with the Krull and covering dimension.

We remind, firstly, basic definitions and notations which will be used in order to define the small inductive dimension. Throughout this paper we suppose that the lattices are finite.

If L is a lattice and $x \in L$, then we write $\uparrow x = \{y \in L : x \leq y\}$ and $x^* = \bigvee \{y \in L : y \wedge x = 0_L\}$. In the case where $x^* \wedge x = 0_L$, that is $x^* = \max \{y \in L : y \wedge x = 0_L\}$, the element x^* is called *pseudocomplement* of x . Every finite distributive lattice is pseudocomplemented. Although the lattices we deal with in this paper are not generally distributive, we will adopt

the asterisk notation and the term “pseudocomplement” in order to have the same terminology as in [2]. We denote by 0_L and 1_L the bottom and the top element of L , respectively.

A subset V of a lattice L is called a *cover* of L if $0_L \notin V$ and $\bigvee V = 1_L$. A cover U of a lattice L is called a *refinement* of a cover V of L , writing $U \succ V$, if for each $u \in U$, there exists $v \in V$ such that $u \leq v$.

A subset B of a lattice L is called a *base* of L if each element of L can be written as the supremum of a subset of B .

We state that in the study [2] the authors define and study the small inductive dimension for regular frames. Also, more results and properties for the dimension ind can be found in [3].

2 The Small Inductive Dimension for Finite Lattices

Based on the notion of the small inductive dimension, ind , which is defined for the class of all regular frames [2], in this section, we study the small inductive dimension ind in the class of finite lattices and we give properties of this dimension.

We mention that we define this notion of dimension for the class of finite lattices because we can see that this meaning is well defined and more friendly for this class. Clearly, we can compute this dimension for regular finite lattices but with our study we realize that this dimension behaves well for an arbitrary finite lattice.

Since the dimension ind is defined inductively for finite lattices, we remind that a *sublattice* of a lattice L is a subset of L , which is a lattice with the same meet and join operations as L . Given a lattice L , every set of the form $\uparrow x$, where $x \in L$, is a sublattice of L .

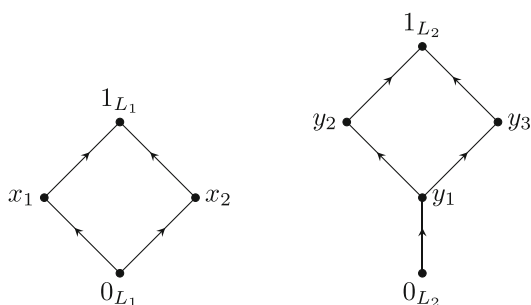
Definition 1 Let L be a finite lattice. The *small inductive dimension*, ind , of L is defined as follows:

1. $\text{ind}(L) = -1$ if and only if $L = \{0_L\}$.
2. $\text{ind}(L) \leq k$, where $k \in \mathbb{N}$, if for every cover V of L , there exists a cover U of L such that U is a refinement of V and $\text{ind}(\uparrow(u^* \vee u)) \leq k - 1$, for every $u \in U$.
3. $\text{ind}(L) = k$, where $k \in \mathbb{N}$, if $\text{ind}(L) \leq k$ and $\text{ind}(L) \not\leq k - 1$.

Analogously to Proposition 4.2 of [2], isomorphic lattices have the same small inductive dimension.

Example 1 We consider the finite lattices represented by the diagrams of Figure 1. We have that $\text{ind}(L_1) = 0$ and $\text{ind}(L_2) = 1$.

Fig. 1 The finite lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$



In the study [2] the authors have proved that for any regular frame L , $\text{ind}(L) \leq k$, where $k \in \mathbb{N}$, if and only if there exists a base B of L such that $\text{ind}(\uparrow(b \vee b^*)) \leq k - 1$, for every $b \in B$.

Proposition 1 *Let L be a finite lattice and $k \in \mathbb{N}$. If there exists a base B of L such that $\text{ind}(\uparrow(b \vee b^*)) \leq k - 1$, for every $b \in B$, then $\text{ind}(L) \leq k$.*

Proof Let V be a cover of L . Since B is a base of L , there exists a cover $B_0 = \{b_{j_1}, \dots, b_{j_m}\} \subseteq B$ of L , where

$$v = \min\{m : \{b_{j_1}, \dots, b_{j_m}\} \text{ is a cover of } L \text{ and } \{b_{j_1}, \dots, b_{j_m}\} > V\}.$$

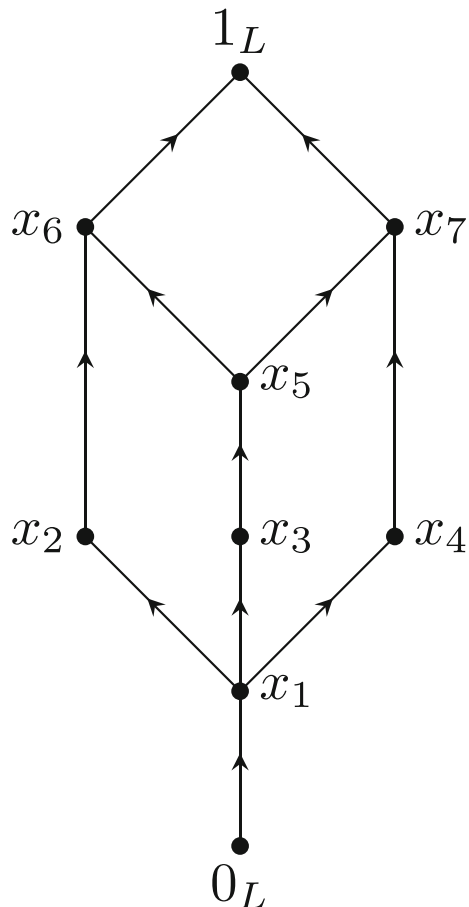
Also, by assumption, $\text{ind}(\uparrow(b \vee b^*)) \leq k - 1$, for every $b \in B_0$. $\square$

However, for the class of finite lattices we observe that the converse of Proposition 1 does not always hold as the lattice of Figure 2 shows.

For the lattice L of Figure 2 we have that $\text{ind}(L) = 1$. Moreover, every base B of L must contain the element x_3 for which

$$\text{ind}(\uparrow(x_3 \vee x_3^*)) = \text{ind}(\uparrow x_3) = 1 \not\leq 1 - 1.$$

Fig. 2 The finite lattice $(L, \leq)$



In the next result a new characterization of the small inductive dimension based on the notion of minimal cover is given.

Definition 2 [8] In any finite lattice L , a cover V of L is called *minimal* if $V \subseteq U$, for every refinement U of V .

Theorem 2 Let L be a finite lattice and $k \in \mathbb{N}$. Then, $\text{ind}(L) \leq k$ if and only if for every minimal cover V of L , $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$, for every $v \in V$.

Proof Let $\text{ind}(L) \leq k$, V be a minimal cover of L and $v \in V$. We shall prove that $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$. Since $\text{ind}(L) \leq k$, there exists a cover U of L , which refines V and $\text{ind}(\uparrow(u^* \vee u)) \leq k - 1$, for every $u \in U$. Since V is a minimal cover of L , we have that $V \subseteq U$ and thus, $v \in U$. Therefore, $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$.

Conversely, we shall prove that $\text{ind}(L) \leq k$. Let W be a cover of L . If $1_L \in W$, then the cover $U = \{1_L\}$ is a refinement of W and

$$\text{ind}(\uparrow(1_L^* \vee 1_L)) = \text{ind}(\uparrow(0_L \vee 1_L)) = \text{ind}(\uparrow 1_L) = \text{ind}(\{1_L\}) = -1.$$

Otherwise, if $1_L \notin W$, by [8, Lemma 3.7], there exists a minimal cover V of L which refines W and $1_L \notin V$. By assumption, $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$, for every $v \in V$. Thus, $\text{ind}(L) \leq k$. $\square$

Proposition 3 Let L be a finite lattice. Then the following are equivalent:

1. $\text{ind}(L) = 0$.
2. For every minimal cover V of L and each $v \in V$, $\{v, v^*\}$ is a cover of L .

Proof By Theorem 2, $\text{ind}(L) = 0$ if and only if for every minimal cover V of L , $\text{ind}(\uparrow(v^* \vee v)) = -1$, for every $v \in V$. Thus, $\text{ind}(L) = 0$ if and only if for every minimal cover V of L , $\uparrow(v^* \vee v) = \{1_L\}$, for every $v \in V$, or equivalently, $\text{ind}(L) = 0$ if and only if $v^* \vee v = 1_L$, for every $v \in V$. $\square$

Theorem 4 For every $k \in \{1, 2, 3, \dots\}$ there exists a finite lattice L such that $\text{ind}(L) = k$ and all $x \in L \setminus \{0_L\}$ have 0_L as pseudocomplement.

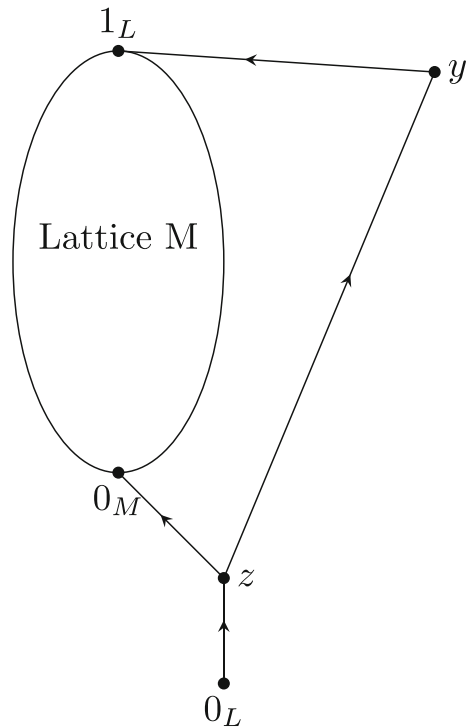
Proof We prove the theorem by induction on k . Firstly, we shall prove that there exists a finite lattice L , in which the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L , with $\text{ind}(L) = 1$. Indeed, if we consider the lattice $(L_2, \leq_2)$ of Example 1, we have the desired result.

We suppose that there exists a finite lattice M , in which the pseudocomplement of every element of $M \setminus \{0_M\}$ is 0_M , with $\text{ind}(M) = k - 1$, where $k > 1$ (induction hypothesis), and we shall prove that there exists a finite lattice L , in which the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L , with $\text{ind}(L) = k$.

We consider the finite lattice $L = \{0_L, z, y\} \cup M$ (as represented in Figure 3), which partial order $\leq$ is defined as follows:

- (1) $z \leq w$, for every $w \in L \setminus \{0_L\}$,
- (2) $w \parallel y$ (i.e. neither $w \leq y$ nor $y \leq w$ holds), for every $w \in M$.

We observe that the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L . We shall prove that $\text{ind}(L) = k$. By Theorem 2, we shall prove that for every minimal cover V of L , $\text{ind}(\uparrow(v \vee v_L^*)) \leq k - 1$, for every $v \in V$, where v_L^* denotes the pseudocomplement of v in L . Let V be a minimal cover of L . According to Figure 3, we study the following cases:

Fig. 3 The lattice $(L, \leq)$ 

(i) For the minimal cover $V = \{0_M, y\}$ of L we have that

$$\text{ind}(\uparrow((0_M)_L^* \vee 0_M)) = \text{ind}(\uparrow(0_L \vee 0_M)) = \text{ind}(\uparrow 0_M) = \text{ind}(M) = k - 1,$$

where $(0_M)_L^*$ denotes the pseudocomplement of 0_M in the lattice L , and

$$\text{ind}(\uparrow(y^* \vee y)) = \text{ind}(\uparrow(0_L \vee y)) = \text{ind}(\uparrow y) = \text{ind}(\{y, 1_L\}) = 0 = 1 - 1 \leq k - 1.$$

(ii) For the minimal covers of M , we have that $\text{ind}(\uparrow(v \vee v_L^*)) \leq k - 1$, for every $v \in V$, where v_L^* denotes the pseudocomplement of v in L .

We state that if V is a minimal cover of M , then it is also a minimal cover of L . First of all, since $\text{ind}(M) > 0$, $1_L \notin V$. Moreover, since V is a cover of M , it is also a cover of L . Let U be a cover of L , which refines V . Then $y \notin U$. (If $y \in U$, since $U \succ V$, there exists $v \in V$ such that $y \leq v$. But the only element for which $y \leq v$ is the element 1_L which does not belong to V .) Since for each $u \in U \setminus \{z\}$, $0_M \leq u$, we have that each $u \in U \setminus \{z\}$ belongs to M . That is, $U \setminus \{z\}$ is a cover of M , which refines V . Since V is minimal cover of M , we have that $V \subseteq U \setminus \{z\}$. Thus, $V \subseteq U$ and therefore, V is a minimal cover of L .

Let $v \in V$, where V is a minimal cover of M . We have that

$$\text{ind}(\uparrow(v \vee v_L^*)) = \text{ind}(\uparrow(v \vee 0_L)) = \text{ind}(\uparrow v).$$

Since $\text{ind}(M) = k - 1$, by Theorem 2 we have that

$$\text{ind}(\uparrow(v \vee v_M^*)) = \text{ind}(\uparrow(v \vee 0_M)) = \text{ind}(\uparrow v) \leq k - 2,$$

where v_M^* denotes the pseudocomplement of v in M . Thus, $\text{ind}(\uparrow(v \vee v_L^*)) \leq k - 1$, for every $v \in V$ and thus, $\text{ind}(L) \leq k$.

Moreover, we observe that there exists the cover $V = \{0_M, y\}$ of L such that for every refinement U of V (that is, $U = \{0_M, y\}$ and $U = \{0_M, y, z\}$), there exists $u = 0_M \in U$ such that

$$\text{ind}(\uparrow((0_M)_L^* \vee 0_M)) = \text{ind}(\uparrow(0_L \vee 0_M)) = \text{ind}(\uparrow 0_M) = \text{ind}(M) = k - 1 \not\leq k - 2.$$

Thus, $\text{ind}(L) \not\leq k - 1$ and so $\text{ind}(L) = k$. $\square$

We also present a sublattice theorem for the small inductive dimension of finite lattices. We remind that a subset S of a lattice L is called *down-set* if for every $x \in L$ with $x \leq s$ for some $s \in S$, then $x \in S$.

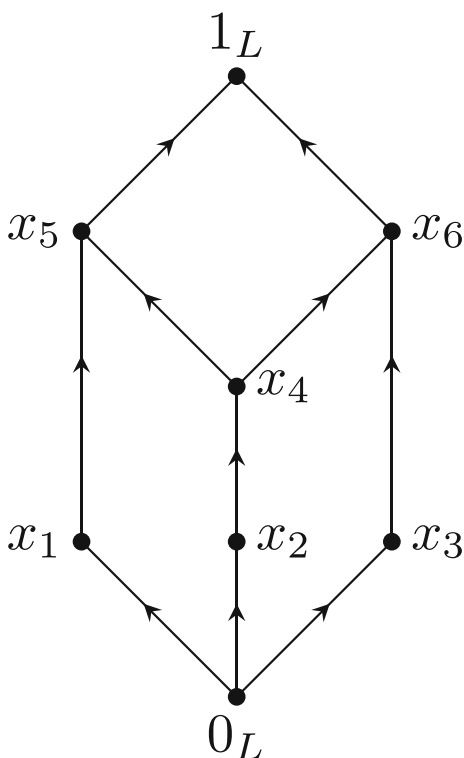
As we observe in Figure 4, a corresponding sublattice theorem does not hold without mentioning further conditions for the sublattice of a finite lattice.

We have that $\text{ind}(L) = 0$ but for the sublattice $M = \{x_2, x_4, x_5, x_6, 1_L\}$ of L we have that $\text{ind}(M) = 1$ and therefore, $\text{ind}(M) \not\leq \text{ind}(L)$.

Proposition 5 *Let L be a finite lattice and M be a sublattice of L such that $1_L \in M$, $M \setminus \{1_L\}$ is a down-set and the pseudocomplement u_M^* of every element u of M is equal to the pseudocomplement u_L^* of u in L . Then, $\text{ind}(M) \leq \text{ind}(L)$.*

Proof If $\text{ind}(L) = -1$, then, clearly, $\text{ind}(M) = -1$. As second case we suppose that $\text{ind}(L) = k$, where $k \in \mathbb{N}$ and we will prove that $\text{ind}(M) \leq k$. Let V be a cover of M . Since $1_L \in M$, V is a cover of L . Since $\text{ind}(L) = k$, there exists a cover U of L , which is a refinement of V and for every $u \in U$, $\text{ind}(\uparrow(u_L^* \vee u)) \leq k - 1$.

Fig. 4 The finite lattice $(L, \leq)$



If $1_L \in V$, then there exists the cover $U = \{1_L\}$ of L , which is a refinement of V and

$$\text{ind}(\uparrow(1_L^* \vee 1_L)) = \text{ind}(\uparrow 1_L) = -1 \leq k - 1.$$

Let $1_L \notin V$. Since $M \setminus \{1_L\}$ is a down set, U is a cover of M . By assumption

$$\text{ind}(\uparrow(u_M^* \vee u)) \leq k - 1.$$

Therefore, in each case $\text{ind}(M) \leq k$. $\square$

3 Properties Concerning Sums and Products

In this section we study properties of the small inductive dimension of different finite lattices. Especially, we study the small inductive dimension of the linear sum and three kinds of products of finite lattices.

Definition 3 [23] The *linear sum* $L_1 \oplus L_2$ of two partially ordered sets $(L_1, \leq_1)$ and $(L_2, \leq_2)$, where $L_1 \cap L_2 = \emptyset$, is the partially ordered set $(L_1 \cup L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$x \leq y \Leftrightarrow \begin{cases} x, y \in L_1 \text{ and } x \leq_1 y \\ x, y \in L_2 \text{ and } x \leq_2 y \\ x \in L_1, y \in L_2. \end{cases}$$

Clearly, if $(L_1, \leq_1)$ and $(L_2, \leq_2)$ are lattices, then $(L_1 \oplus L_2, \leq)$ is also a lattice.

Remark 1 (1) The lattices of Figures 1, 5 and 6 verifies the claim

$$\text{ind}(L_1 \oplus L_3) \neq \text{ind}(L_3 \oplus L_1).$$

We have that $\text{ind}(L_3) = \text{ind}(L_1) = 0$. Also, $\text{ind}(L_3 \oplus L_1) = 1$ and $\text{ind}(L_1 \oplus L_3) = 0$. Thus, $\text{ind}(L_1 \oplus L_3) \neq \text{ind}(L_3 \oplus L_1)$.

(2) The lattices of Figures 5, 7 and 8 verifies that the assertion

$$\text{ind}(L_3 \oplus L_4) \leq \text{ind}(L_3) + \text{ind}(L_4) + 1$$

does not hold for all finite lattices L_3 and L_4 .

We have that $\text{ind}(L_3) = \text{ind}(L_4) = 0$ and $\text{ind}(L_3 \oplus L_4) = 2$. Thus,

$$\text{ind}(L_3 \oplus L_4) \not\leq \text{ind}(L_3) + \text{ind}(L_4) + 1.$$

Proposition 6 Let L_1 and L_2 be two finite lattices such that the pseudocomplement of every element of $L_2 \setminus \{0_{L_2}\}$ is 0_{L_2} . Then $\text{ind}(L_2) = \text{ind}(L_1 \oplus L_2)$.

Fig. 5 The finite lattice $(L_3, \leq_3)$



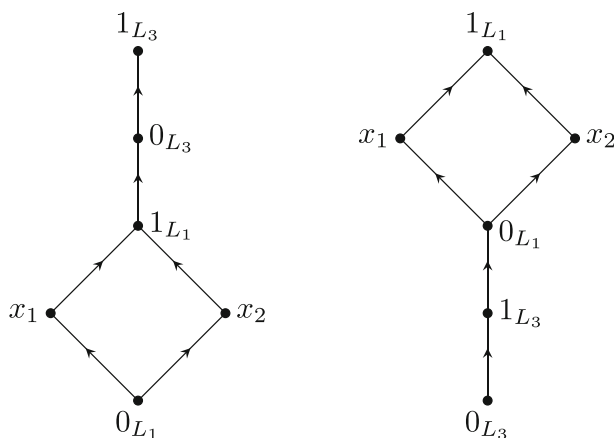


Fig. 6 The finite lattices $(L_1 \oplus L_3, \leq)$ and $(L_3 \oplus L_1, \leq^*)$

Proof We prove that one dimension is less than or equal to the other dimension. Firstly, we shall prove that $\text{ind}(L_2) \leq \text{ind}(L_1 \oplus L_2)$. Let $\text{ind}(L_1 \oplus L_2) = k$, where $k \in \mathbb{N}$. We shall prove that $\text{ind}(L_2) \leq k$. Let V be a minimal cover of L_2 and $v \in V$. By [8] V is a minimal cover of $L_1 \oplus L_2$. Since $\text{ind}(L_1 \oplus L_2) = k$, by Theorem 2

$$\text{ind}(\uparrow(v \vee v_{L_1 \oplus L_2}^*)) \leq k - 1,$$

where $v_{L_1 \oplus L_2}^*$ denotes the pseudocomplement of v in $L_1 \oplus L_2$. Since $v \in L_2$, $v_{L_1 \oplus L_2}^* = 0_{L_1 \oplus L_2}$ and thus, $\text{ind}(\uparrow v) \leq k - 1$. Moreover, in the lattice L_2 we have that $v_{L_2}^* = 0_{L_2}$, where $v_{L_2}^*$ denotes the pseudocomplement of v in L_2 . Therefore,

$$\text{ind}(\uparrow(v \vee v_{L_2}^*)) = \text{ind}(\uparrow v) \leq k - 1$$

and thus, $\text{ind}(L_2) \leq k$.

Now, we shall prove that $\text{ind}(L_1 \oplus L_2) \leq \text{ind}(L_2)$. Let $\text{ind}(L_2) = k$, where $k \in \mathbb{N}$. We shall prove that $\text{ind}(L_1 \oplus L_2) \leq k$. Let V be a minimal cover of $L_1 \oplus L_2$ and $v \in V$. Then

Fig. 7 The finite lattice $(L_4, \leq_4)$

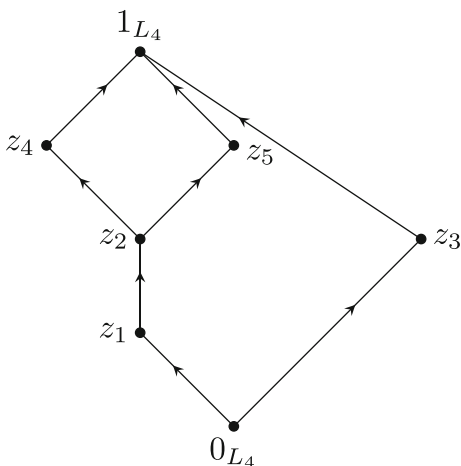
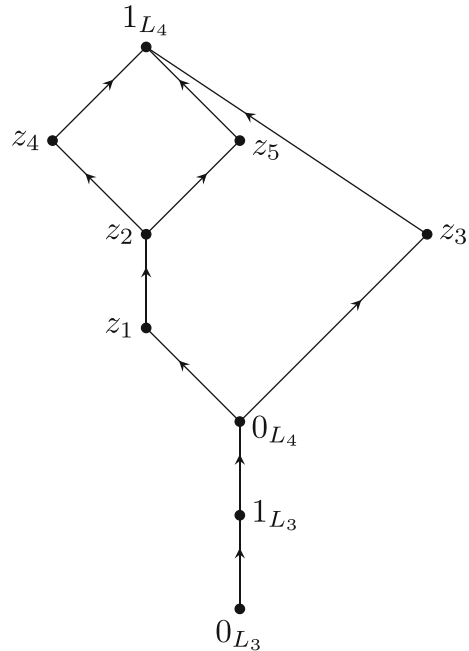


Fig. 8 The finite lattice $(L_3 \oplus L_4, \leq)$



V is a minimal cover of L_2 . Since $\text{ind}(L_2) = k$, by Theorem 2

$$\text{ind}(\uparrow(v \vee v_{L_2}^*)) \leq k - 1$$

and by assumption $\text{ind}(\uparrow v) \leq k - 1$. Moreover, $v_{L_1 \oplus L_2}^* = 0_{L_1 \oplus L_2}$ and thus,

$$\text{ind}(\uparrow v \vee v_{L_1 \oplus L_2}^*) = \text{ind}(\uparrow v) \leq k - 1.$$

Therefore, $\text{ind}(L_1 \oplus L_2) \leq k$. $\square$

Definition 4 The *Cartesian product* of two partially ordered sets $(L_1, \leq_1)$ and $(L_2, \leq_2)$ is the partially ordered set $(L_1 \times L_2, \leq)$, where the relation $\leq$ is defined as follows: $(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow x_1 \leq_1 x_2$ and $y_1 \leq_2 y_2$.

If $(L_1, \leq_1)$ and $(L_2, \leq_2)$ are finite lattices, then $(L_1 \times L_2, \leq)$ is also a finite lattice.

The following lemma can easily be proved and thus, we omit the proof.

Lemma 7 If L_1 and L_2 are finite lattices and $(x, y) \in L_1 \times L_2$, then the following are satisfied:

1. $\uparrow(x, y) = \uparrow x \times \uparrow y$,
2. $(x, y)^* = (x^*, y^*)$.

Theorem 8 Let L_1 and L_2 be finite lattices. Then,

$$\text{ind}(L_1 \times L_2) = \max\{\text{ind}(L_1), \text{ind}(L_2)\}.$$

Proof In order to prove the inequality $\text{ind}(L_1 \times L_2) \leq \max\{\text{ind}(L_1), \text{ind}(L_2)\}$, we shall apply induction with respect to the number

$$k(L_1, L_2) = \max\{\text{ind}(L_1), \text{ind}(L_2)\}.$$

If $k(L_1, L_2) = -1$, then our inequality holds. Moreover, if $\text{ind}(L_1) = -1$ and $\text{ind}(L_2) \geq 0$ (respectively, $\text{ind}(L_2) = -1$ and $\text{ind}(L_1) \geq 0$), then the lattices $L_1 \times L_2$ and L_2 (respectively, $L_1 \times L_2$ and L_1) are isomorphic and therefore, $\text{ind}(L_1 \times L_2) = \text{ind}(L_2)$ (respectively, $\text{ind}(L_1 \times L_2) = \text{ind}(L_1)$).

We assume that the inequality holds for every pair of finite lattices L_1 and L_2 with $k(L_1, L_2) < k$, where $k \geq 0$, and we consider finite lattices L_1 and L_2 such that $\text{ind}(L_1) = n \geq 0$, $\text{ind}(L_2) = m \geq 0$ and $\max\{n, m\} = k$. We shall prove that $\text{ind}(L_1 \times L_2) \leq k$.

Let V be a minimal cover of $L_1 \times L_2$. Then for every $(a, b) \in V$ we have either $a = 0_{L_1}$ or $b = 0_{L_2}$ (see [20]). Thus, V is of the form

$$V = \{(a_1, 0_{L_2}), \dots, (a_s, 0_{L_2}), (0_{L_1}, b_1), \dots, (0_{L_1}, b_t)\}.$$

Then, we observe that the sets $V_1 = \{a_1, \dots, a_s\}$ and $V_2 = \{b_1, \dots, b_t\}$ are minimal covers of the lattices L_1 and L_2 , respectively.

Since $\text{ind}(L_1) = n$, we have that $\text{ind}(\uparrow(a_i^* \vee a_i)) \leq n - 1$, for each $a_i \in V_1, i \in \{1, \dots, s\}$. Also, since $\text{ind}(L_2) = m$, we have that $\text{ind}(\uparrow(b_j^* \vee b_j)) \leq m - 1$, for each $b_j \in V_2, j \in \{1, \dots, t\}$.

Let $v \in V$. It suffices to prove that

$$\text{ind}(\uparrow(v^* \vee v)) \leq \max\{n, m\} - 1 = \max\{n - 1, m - 1\}.$$

We study the following cases:

(A) Let $i \in \{1, \dots, s\}$ such that $v = (a_i, 0_{L_2})$. In this case we have that $v^* = (a_i^*, 1_{L_2})$ and thus,

$$\text{ind}(\uparrow(v^* \vee v)) = \text{ind}(\uparrow(a_i^* \vee a_i, 1_{L_2})).$$

Then by Lemma 7, $\uparrow(a_i^* \vee a_i, 1_{L_2}) = \uparrow(a_i^* \vee a_i) \times \uparrow 1_{L_2}$ and thus we have that

$$\text{ind}(\uparrow(a_i^* \vee a_i, 1_{L_2})) = \text{ind}(\uparrow(a_i^* \vee a_i) \times \uparrow 1_{L_2}).$$

Since $\text{ind}(\uparrow(a_i^* \vee a_i)) \leq n - 1$ and $\text{ind}(\uparrow 1_{L_2}) = \text{ind}(\{1_{L_2}\}) = -1 \leq m - 1$,

$$k(\uparrow(a_i^* \vee a_i), \uparrow 1_{L_2}) \leq \max\{n - 1, -1\} = n - 1 < n \leq k.$$

Therefore, by the induction hypothesis, we have that

$$\begin{aligned} \text{ind}(\uparrow(a_i^* \vee a_i) \times \uparrow 1_{L_2}) &\leq \max\{\text{ind}(\uparrow(a_i^* \vee a_i)), \text{ind}(\uparrow 1_{L_2})\} \\ &\leq \max\{n - 1, m - 1\} = \max\{n, m\} - 1. \end{aligned}$$

Thus, $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$.

(B) Let $j \in \{1, \dots, t\}$ such that $v = (0_{L_1}, b_j)$. In this case we have that $v^* = (1_{L_1}, b_j^*)$ and thus,

$$\text{ind}(\uparrow(v^* \vee v)) = \text{ind}(\uparrow(1_{L_1}, b_j^* \vee b_j)).$$

As in case **(A)**, we have that

$$\text{ind}(\uparrow(1_{L_1}, b_j^* \vee b_j)) = \text{ind}(\uparrow 1_{L_1} \times \uparrow(b_j^* \vee b_j)).$$

Since $\text{ind}(\uparrow 1_{L_1}) = \text{ind}(\{1_{L_1}\}) = -1 \leq n - 1$ and $\text{ind}(\uparrow(b_j^* \vee b_j)) \leq m - 1$,

$$k(\uparrow 1_{L_1}, \uparrow(b_j^* \vee b_j)) \leq \max\{-1, m - 1\} = m - 1 < m \leq k.$$

Therefore, by the induction hypothesis, we have that

$$\begin{aligned} \text{ind}(\uparrow 1_{L_1} \times \uparrow(b_j^* \vee b_j)) &\leq \max\{\text{ind}(\uparrow 1_{L_1}), \text{ind}(\uparrow(b_j^* \vee b_j))\} \\ &\leq \max\{n - 1, m - 1\} = \max\{n, m\} - 1. \end{aligned}$$

Thus, $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$.

Considering the above cases, we have that $\text{ind}(L_1 \times L_2) \leq k$.

It remains to prove the inequality

$$\max\{\text{ind}(L_1), \text{ind}(L_2)\} \leq \text{ind}(L_1 \times L_2).$$

If $\text{ind}(L_1 \times L_2) = -1$, then $L_1 \times L_2 = \{1_{L_1 \times L_2}\}$ and thus, $L_1 = \{1_{L_1}\}$ and $L_2 = \{1_{L_2}\}$ and the inequality holds.

We suppose that $\text{ind}(L_1 \times L_2) = k$, where $k \in \mathbb{N}$ and we shall prove that $\text{ind}(L_1) \leq k$ and $\text{ind}(L_2) \leq k$. Let $V_1 = \{a_1, \dots, a_s\}$ and $V_2 = \{b_1, \dots, b_t\}$ be minimal covers of L_1 and L_2 , respectively. We consider the set

$$W = \{(a_1, 0_{L_2}), \dots, (a_s, 0_{L_2}), (0_{L_1}, b_1), \dots, (0_{L_1}, b_t)\}.$$

Then W is a minimal cover of $L_1 \times L_2$. Indeed, let $C = \{(x_1, y_1), \dots, (x_p, y_p)\}$ be a cover of $L_1 \times L_2$ which refines W . Then each $(x_i, y_i) \in C$, $i \in \{1, \dots, p\}$, is of the form $(x_i, 0_{L_2})$ and $(0_{L_1}, y_i)$. Moreover, since $\{x_1, \dots, x_p\}$ and $\{y_1, \dots, y_p\}$ are covers of L_1 and L_2 , which refine V_1 and V_2 respectively, we have that $V_1 \subseteq \{x_1, \dots, x_p\}$ and $V_2 \subseteq \{y_1, \dots, y_p\}$. Therefore, $W \subseteq C$.

Since $\text{ind}(L_1 \times L_2) = k$, by Theorem 2 we have that

$$\text{ind}(\uparrow((a_i, 0_{L_2}) \vee (a_i, 0_{L_2})^*)) \leq k - 1,$$

for every $i \in \{1, \dots, s\}$ and

$$\text{ind}(\uparrow((0_{L_1}, b_j) \vee (0_{L_1}, b_j)^*)) \leq k - 1,$$

for every $j \in \{1, \dots, t\}$. Then

$$\begin{aligned} \text{ind}(\uparrow(a_i \vee a_i^*)) &= \text{ind}(\uparrow(a_i \vee a_i^*) \times \{1_{L_2}\}) \\ &= \text{ind}(\uparrow(a_i \vee a_i^*, 1_{L_2})) \\ &= \text{ind}(\uparrow((a_i, 0_{L_2}) \vee (a_i, 0_{L_2})^*)) \\ &\leq k - 1, \end{aligned}$$

for every $i \in \{1, \dots, s\}$ and

$$\begin{aligned} \text{ind}(\uparrow(b_j \vee b_j^*)) &= \text{ind}(\{1_{L_1}\} \times \uparrow(b_j \vee b_j^*)) \\ &= \text{ind}(\uparrow(1_{L_1}, b_j \vee b_j^*)) \\ &= \text{ind}(\uparrow((0_{L_1}, b_j) \vee (0_{L_1}, b_j)^*)) \\ &\leq k - 1, \end{aligned}$$

for every $j \in \{1, \dots, t\}$. Therefore, $\text{ind}(L_1) \leq k_1$ and $\text{ind}(L_2) \leq k$. $\square$

Corollary 9 Let L_1 and L_2 be finite lattices. Then,

$$\text{ind}(L_1 \times L_2) < \text{ind}(L_1) + \text{ind}(L_2) + 1.$$

Definition 5 For two lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$ the *lexicographic product* $L_1 \diamond L_2$ is the lattice $(L_1 \times L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow \begin{cases} x_1 <_1 x_2 \text{ or} \\ x_1 = x_2 \text{ and } y_1 \leq_2 y_2. \end{cases}$$

By Definition 4.8 of [8] for every $(x, y), (x', y') \in L_1 \diamond L_2$ we have that

$$(x, y) \wedge (x', y') = \begin{cases} (x, y), & \text{if } x < x' \\ (x', y'), & \text{if } x' < x \\ (x, y \wedge y'), & \text{if } x = x' \\ (x \wedge x', 1_{L_2}), & \text{if } x \parallel x' \end{cases}$$

and

$$(x, y) \vee (x', y') = \begin{cases} (x, y), & \text{if } x' < x \\ (x', y'), & \text{if } x < x' \\ (x, y \vee y'), & \text{if } x = x' \\ (x \vee x', 0_{L_2}), & \text{if } x \parallel x'. \end{cases}$$

Remark 2 The lattices of Figures 5, 7 and 9 verifies that the inequality

$$\text{ind}(L_3 \diamond L_4) \leq \text{ind}(L_3) + \text{ind}(L_4) + 1$$

does not hold for all finite lattices L_3 and L_4 .

For the finite lattices $(L_3, \leq_3)$ and $(L_4, \leq_4)$ of Figures 5 and 7, respectively we have that $\text{ind}(L_3) = \text{ind}(L_4) = 0$. The finite lattice $(L_3 \diamond L_4, \leq)$ is represented by the diagram of Figure 9.

We have that $\text{ind}(L_3 \diamond L_4) = 2$. Thus, the inequality

$$\text{ind}(L_3 \diamond L_4) \leq \text{ind}(L_3) + \text{ind}(L_4) + 1$$

does not hold.

Proposition 10 Let U be a cover of $L_1 \diamond L_2$ that does not contain $(1_{L_1}, 1_{L_2})$. Then, there exists a cover $W = \{(a_1, b_1), \dots, (a_m, b_m)\}$ of $L_1 \diamond L_2$ such that $W \subseteq U$, $a_1 = \dots = a_m = 1_{L_1}$, $b_1 \vee \dots \vee b_m = 1_{L_2}$ and $b_i \neq 1_{L_2}$, for every $i = 1, \dots, m$.

Proof Let $U = \{(x_1, y_1), \dots, (x_p, y_p)\}$ be a cover of $L_1 \diamond L_2$ that does not contain $(1_{L_1}, 1_{L_2})$.

If $x_i \neq 1_{L_1}$, for every $i = 1, \dots, p$, then $(1_{L_1}, 0_{L_2})$ is an upper bound of U , which is a contradiction (since U is a cover of $L_1 \diamond L_2$).

Let $\{i : x_i = 1_{L_1}\} = \{i_1, \dots, i_m\}$. Hence $x_i \neq 1_{L_1}$ for every $i \notin \{i_1, \dots, i_m\}$. Then, $y_i \neq 1_{L_2}$, $i = i_1, \dots, i_m$. In the case where $m = p$, then U is the required cover. If $y_{i_1} \vee \dots \vee y_{i_m} = z \neq 1_{L_2}$, then $(1_{L_1}, z)$ is an upper bound of U , which is a contradiction. Therefore, $y_{i_1} \vee \dots \vee y_{i_m} = 1_{L_2}$. The cover $W = \{(x_{i_1}, y_{i_1}), \dots, (x_{i_m}, y_{i_m})\}$ is the required cover.

Corollary 11 Let $U = \{(a_1, b_1), \dots, (a_m, b_m)\}$ be a minimal cover of $L_1 \diamond L_2$ that does not contain $(1_{L_1}, 1_{L_2})$. Then, $a_1 = \dots = a_m = 1_{L_1}$, $b_1 \vee \dots \vee b_m = 1_{L_2}$ and $b_i \neq 1_{L_2}$, for every $i = 1, \dots, m$. Moreover, the cover $\{b_1, \dots, b_m\}$ of L_2 is minimal.

Proposition 12 Let L_1 and L_2 be two finite lattices such that the pseudocomplement of every element of $L_2 \setminus \{0_{L_2}\}$ is 0_{L_2} . Then $\text{ind}(L_1 \diamond L_2) \leq \text{ind}(L_2)$.

Proof Let $\text{ind}(L_2) = k$, where $k \in \mathbb{N}$. We prove that $\text{ind}(L_1 \diamond L_2) \leq k$. If every cover of $L_1 \diamond L_2$ contains $(1_{L_1}, 1_{L_2})$ (that is $\{(1_{L_1}, 1_{L_2})\}$ is a minimal cover of $L_1 \diamond L_2$), then $\text{ind}(L_1 \diamond L_2) = 0$ and the inequality holds.

We suppose $L_1 \diamond L_2$ has at least one cover that does not contain $(1_{L_1}, 1_{L_2})$. Let $V = \{(a_1, b_1), \dots, (a_m, b_m)\}$ be a minimal cover of $L_1 \diamond L_2$ (see Lemma 3.7 of [11]) such that

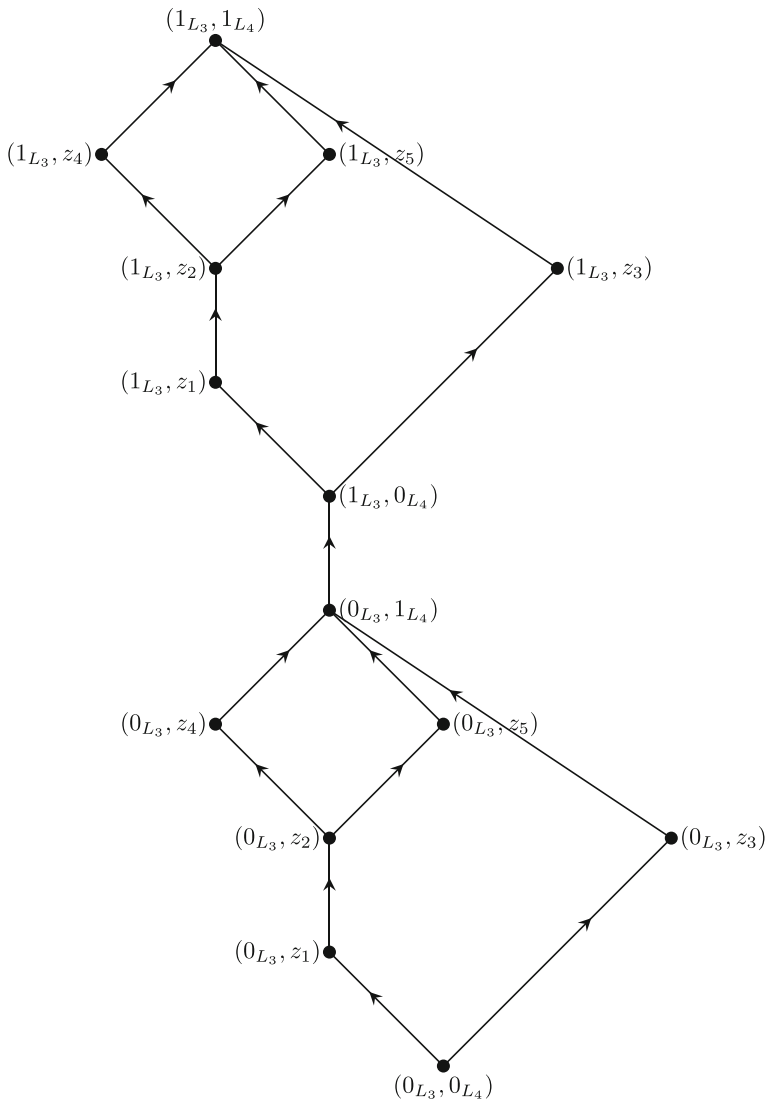


Fig. 9 The finite lattice $(L_3 \diamond L_4, \leq)$

$a_1 = \dots = a_m = 1_{L_1}$, $b_1 \vee \dots \vee b_m = 1_{L_2}$, $b_i \neq 1_{L_2}$, for every $i = 1, \dots, m$ and $\{b_1, \dots, b_m\}$ is a minimal cover of L_2 (Corollary 11). That is,

$$V = \{(1_{L_1}, b_1), \dots, (1_{L_1}, b_m)\}.$$

By Theorem 2 we shall prove that for every $i \in \{1, \dots, m\}$,

$$\text{ind}(\uparrow((1_{L_1}, b_i) \vee (1_{L_1}, b_i)^*)) \leq k - 1$$

and since $(1_{L_1}, b_i) = (0_{L_1}, 0_{L_2})$, for every $i \in \{1, \dots, m\}$, it suffices to prove that for every $i \in \{1, \dots, m\}$,

$$\text{ind}(\uparrow(1_{L_1}, b_i)) \leq k - 1.$$

Let $i \in \{1, \dots, m\}$. Since $\{b_1, \dots, b_m\}$ is a minimal cover of L_2 and $\text{ind}(L_2) = k$, we have that

$$\text{ind}(\uparrow(b_i \vee b_i^*)) = \text{ind}(\uparrow b_i) \leq k - 1.$$

Since $\uparrow(1_{L_1}, b_i) = \uparrow 1_{L_1} \times \uparrow b_i$, we have that

$$\text{ind}(\uparrow(1_{L_1}, b_i)) = \text{ind}(\{1_{L_1}\} \times \uparrow b_i) = \text{ind}(\uparrow b_i) = k - 1.$$

□

Remark 3 We also have that the small inductive dimension of the Cartesian product $L_1 \times L_2$ of two finite lattices L_1 and L_2 differs from the small inductive dimension of the corresponding lexicographic product.

(1) We consider the finite lattices L_1 and L_3 , which are represented in the diagrams of Figures 1 and 5, respectively. We have that $\text{ind}(L_1) = \text{ind}(L_3) = 0$. Also, by Theorem 8 we have $\text{ind}(L_3 \times L_1) = 0$. The lexicographic product $L_3 \diamond L_1$ is represented in Figure 10.

We observe that $\text{ind}(L_3 \diamond L_1) = 1 \neq \text{ind}(L_3 \times L_1) = 0$.

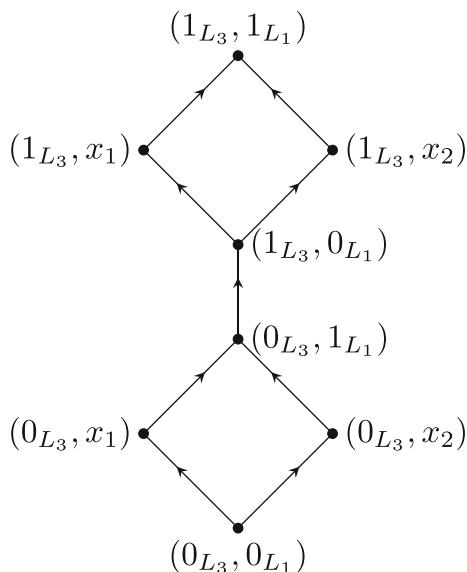
(2) We consider the finite lattice L_2 and L_3 , which are represented in the diagrams of Figures 1 and 5, respectively. We have that $\text{ind}(L_2) = 1$ and $\text{ind}(L_3) = 0$. Also, by Theorem 8, we have that $\text{ind}(L_2 \times L_3) = 1$. Moreover, since the cover $\{(1_{L_2}, 1_{L_3})\}$ is a minimal cover of $L_2 \diamond L_3$, we have that $\text{ind}(L_2 \diamond L_3) = 0$.

Indeed, we shall prove that every cover U of $L_2 \diamond L_3$ contains the element $(1_{L_2}, 1_{L_3})$. If we suppose that there exists a cover $U = \{(a_1, b_1), \dots, (a_m, b_m)\}$ that does not contain $(1_{L_2}, 1_{L_3})$, then the least upper bound of U is the element $(1_{L_2}, 0_{L_3})$ and thus, U is not a cover of $L_2 \diamond L_3$.

Definition 6 [24, 25] The *rectangular product* of two finite lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$ is the lattice $(L_1 \square L_2, \leq)$, where

$$L_1 \square L_2 = \{(x, y) \in L_1 \times L_2 : x \neq 0_{L_1} \text{ and } y \neq 0_{L_2}\} \cup \{(0_{L_1}, 0_{L_2})\}$$

Fig. 10 The finite lattice $(L_3 \diamond L_1, \leq)$



and the relation $\leq$ is defined as follows:

$$(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow x_1 \leq_1 x_2 \text{ and } y_1 \leq_2 y_2.$$

It is known that for every $(x_1, y_1), (x_2, y_2) \in L_1 \square L_2$ (see [8]),

$$(x_1, y_1) \wedge (x_2, y_2) = \begin{cases} (x_1 \wedge x_2, y_1 \wedge y_2), & \text{if } x_1 \wedge x_2 \neq 0_{L_1} \text{ and } y_1 \wedge y_2 \neq 0_{L_2} \\ (0_{L_1}, 0_{L_2}), & \text{otherwise} \end{cases}$$

and $(x_1, y_1) \vee (x_2, y_2) = (x_1 \vee x_2, y_1 \vee y_2)$.

In the case of rectangular product, in Remark 4, we observe that the product theorem of the form $\text{ind}(L_1 \square L_2) \leq \text{ind}(L_1) + \text{ind}(L_2)$ does not hold for all finite lattices L_1 and L_2 , with $\text{ind}(L_1) \geq 0$ and $\text{ind}(L_2) \geq 0$, respectively. On the other side, there exist finite lattices L_1 and L_2 , with $\text{ind}(L_1) \geq 0$ and $\text{ind}(L_2) \geq 0$ for which the product theorem of the form $\text{ind}(L_1 \square L_2) \leq \text{ind}(L_1) + \text{ind}(L_2) + 1$ holds. These two facts led to a natural question, whether there exists a kind of product theorem for the rectangular product.

Remark 4 (1) The lattices of Figure 11 and Figure 12 verifies that the inequality

$$\text{ind}(L_1 \square L_2) \leq \text{ind}(L_1) + \text{ind}(L_2)$$

does not hold for all finite lattices L_1 and L_2 .

We have that $\text{ind}(L_1) = \text{ind}(L_2) = 0$ but $\text{ind}(L_1 \square L_2) = 1$.

(2) We consider the finite lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$ of Figure 1. Then $\text{ind}(L_1) = 0$ and $\text{ind}(L_2) = 1$. The rectangular product $L_2 \square L_1$ is represented by the diagram of Figure 13.

We have that $\text{ind}(L_2 \square L_1) = 0 \leq \text{ind}(L_2) + \text{ind}(L_1) + 1$.

In order to study the relation of the rectangular product of two finite lattices with the small inductive dimension, it is necessary to give an investigation of the following facts, which will be very useful.

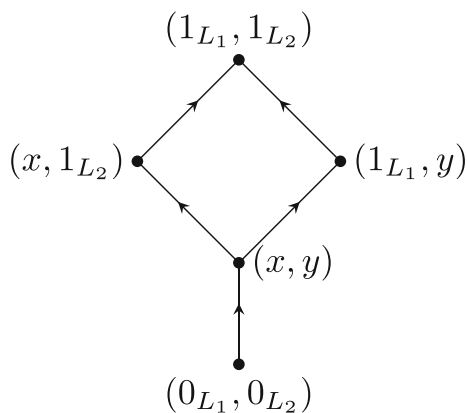
Proposition 13 Let L_1 and L_2 be two finite lattices and $(a, b) \in L_1 \square L_2$. Then

$$(a, b)^* = \begin{cases} (a^*, b^*) = (0_{L_1}, 0_{L_2}), & \text{if } a^* = 0_{L_1} \text{ and } b^* = 0_{L_2} \\ (1_{L_1}, 1_{L_2}), & \text{if } a^* \neq 0_{L_1} \text{ and } b^* \neq 0_{L_2} \\ (1_{L_1}, b^*), & \text{if } a^* = 0_{L_1} \text{ and } b^* \neq 0_{L_2} \\ (a^*, 1_{L_2}), & \text{if } a^* \neq 0_{L_1} \text{ and } b^* = 0_{L_2}. \end{cases}$$

Fig. 11 The finite lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$



Fig. 12 The finite lattice $(L_1 \square L_2, \leq)$



Proof We have that

$$\begin{aligned} (a, b)^* &= \bigvee \{(x, y) \in L_1 \square L_2 : (a, b) \wedge (x, y) = (0_{L_1}, 0_{L_2})\} \\ &= \bigvee \{(x, y) \in L_1 \square L_2 : a \wedge x = 0_{L_1} \text{ or } b \wedge y = 0_{L_2}\}. \end{aligned}$$

Let $M = \{(x, y) \in L_1 \square L_2 : a \wedge x = 0_{L_1} \text{ or } b \wedge y = 0_{L_2}\}$. Then we have the following cases:

(i) If $a^* = 0_{L_1}$ and $b^* = 0_{L_2}$, then $M = \{(0_{L_1}, 0_{L_2})\}$ and therefore

$$(a, b)^* = (a^*, b^*) = (0_{L_1}, 0_{L_2}).$$

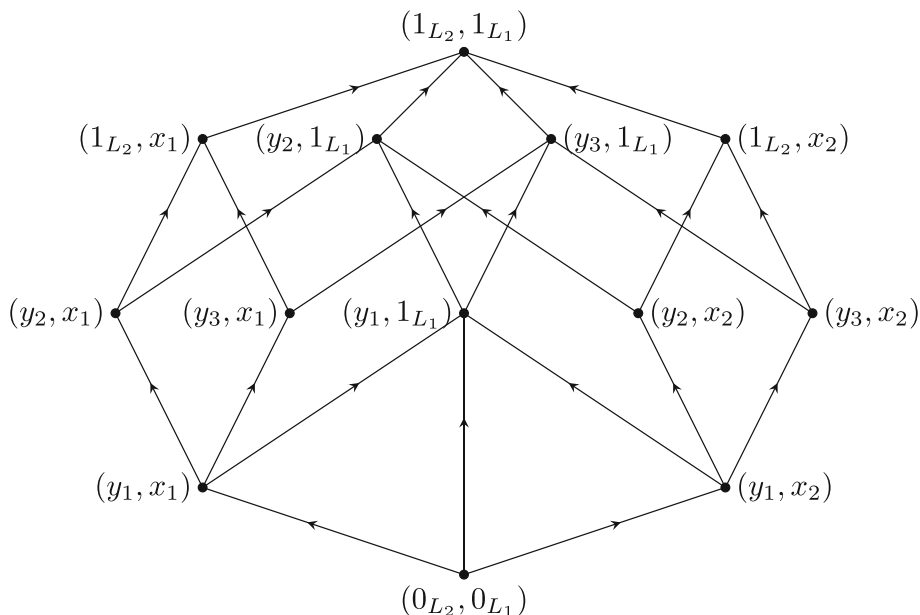


Fig. 13 The finite lattice $(L_2 \square L_1, \leq)$

(ii) If $a^* \neq 0_{L_1}$ and $b^* \neq 0_{L_2}$, then there exist $x \neq 0_{L_1}$ and $y \neq 0_{L_2}$ such that $a \wedge x = 0_{L_1}$ and $b \wedge y = 0_{L_2}$. Therefore $(x, 1_{L_2}) \in M$ and $(1_{L_1}, y) \in M$. Thus, $(1_{L_1}, 1_{L_2})$ is the least upper bound of M and hence

$$(a, b)^* = (1_{L_1}, 1_{L_2}).$$

(iii) If $a^* = 0_{L_1}$ and $b^* \neq 0_{L_2}$, then $a \wedge x = 0_{L_1} \Leftrightarrow x = 0_{L_1}$ and there exists $y \neq 0_{L_2}$ such that $b \wedge y = 0_{L_2}$. Therefore $M = \{(x, y) \in L_1 \square L_2 : b \wedge y = 0_{L_2}\}$. Hence

$$(a, b)^* = (1_{L_1}, b^*).$$

(iv) If $b^* = 0_{L_2}$ and $a^* \neq 0_{L_1}$, then $b \wedge y = 0_{L_2} \Leftrightarrow y = 0_{L_2}$ and there exists $x \neq 0_{L_1}$ such that $a \wedge x = 0_{L_1}$. Therefore $M = \{(x, y) \in L_1 \square L_2 : a \wedge x = 0_{L_1}\}$. Hence

$$(a, b)^* = (a^*, 1_{L_2}).$$

□

Lemma 14 Let L_1 and L_2 be two finite lattices and

$$\{(1_{L_1}, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$$

be a minimal cover of $L_1 \square L_2$, where $m \geq 2$ and $a_i \neq 1_{L_1}$, for every $i \in \{2, \dots, m\}$. Then $\{1_{L_1}\}$ is a minimal cover of L_1 .

Proof By Lemma 4.18 of [8] we have that $b_i \neq b_j$, for every $i, j \in \{1, 2, \dots, m\}$ with $i \neq j$. Let $\{x_1, \dots, x_p\}$ be a minimal cover of L_1 . Then the set

$$\{(x_1, b_1), (x_2, b_1), \dots, (x_p, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$$

is a cover of $L_1 \square L_2$ which refines the minimal cover

$$\{(1_{L_1}, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$$

of $L_1 \square L_2$. Since $\{(1_{L_1}, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$ is minimal, we have that $x_i = 1_{L_1}$, for some $i \in \{1, \dots, p\}$. Since $\{x_1, \dots, x_p\}$ is minimal with $1_{L_1} \in \{x_1, \dots, x_p\}$, we have that $x_1 = \dots = x_p = 1_{L_1}$. Therefore, $\{1_{L_1}\}$ is a minimal cover of L_1 .

Lemma 15 Let L_1 and L_2 be two finite lattices and

$$\{(a_1, 1_{L_2}), (a_2, b_2), \dots, (a_m, b_m)\}$$

be a minimal cover of $L_1 \square L_2$, where $m \geq 2$ and $b_i \neq 1_{L_2}$, for every $i \in \{2, \dots, m\}$. Then $\{1_{L_2}\}$ is a minimal cover of L_2 .

Proof The proof is similar to the proof of Lemma 14. □

Lemma 16 Let L_1 and L_2 be two finite lattices and $\{(a_1, 1_{L_2}), (1_{L_1}, b_2)\}$ be a minimal cover of $L_1 \square L_2$, where $a_1 \neq 1_{L_1}$ and $b_2 \neq 1_{L_2}$. Then $\{1_{L_1}\}$ and $\{1_{L_2}\}$ are minimal covers of L_1 and L_2 , respectively.

Proof Also this proof is similar to the proof of Lemma 14. □

As we have observed in Remark 4, our goal is to provide a kind of product theorem for the case of rectangular product. Therefore, based on the above lemmas, we investigate the corresponding rectangular product theorem. However, in order to succeed this result, we

study the small inductive dimension of the rectangular product of two finite lattices L_1 and L_2 in parts.

By Theorem 2 we have a characterization of the small inductive dimension using minimal covers. Thus, in order to study the small inductive dimension of the rectangular product of two finite lattices L_1 and L_2 it suffices to study propositions related to the minimal covers of $L_1 \square L_2$. This is succeeded in the following four propositions. In Theorem 21 we have the desired result.

Proposition 17 *Let L_1 and L_2 be two finite lattices. For any minimal cover $V = \{(a_1, b_1), (a_2, b_2)\}$ of $L_1 \square L_2$, which consists of two elements, the following inequality holds*

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{\text{ind}(L_1) - 1, \text{ind}(L_2) - 1, 0\},$$

for every $i \in \{1, 2\}$.

Proof We suppose that $\text{ind}(L_1) = k_1 \geq 0$, $\text{ind}(L_2) = k_2 \geq 0$. We shall prove that

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\},$$

for every $i \in \{1, 2\}$. Firstly, by Proposition 13, we observe that for the elements $(a_i, b_i) \in V$, $i \in \{1, 2\}$, one of the following equations is true:

- 1) $(a_i, b_i) \vee (a_i, b_i)^* = (a_i, b_i) \vee (a_i^*, 1_{L_2}) = (a_i \vee a_i^*, 1_{L_2})$,
- 2) $(a_i, b_i) \vee (a_i, b_i)^* = (a_i, b_i) \vee (1_{L_1}, b_i^*) = (1_{L_1}, b_i \vee b_i^*)$,
- 3) $(a_i, b_i) \vee (a_i, b_i)^* = (a_i, b_i) \vee (0_{L_1}, 0_{L_2}) = (a_i, b_i)$,
- 4) $(a_i, b_i) \vee (a_i, b_i)^* = (a_i, b_i) \vee (1_{L_1}, 1_{L_2}) = (1_{L_1}, 1_{L_2})$.

Thus, by Lemma 7 and Theorem 8, in each of these cases we have that

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) = \max\{\text{ind}(\uparrow(a_i \vee a_i^*)), \text{ind}(\uparrow(b_i \vee b_i^*))\}, \quad i \in \{1, 2\}.$$

We have the following cases:

- (i) $1_{L_1} \notin \{a_1, a_2\}$ and $1_{L_2} \notin \{b_1, b_2\}$,
- (ii) $1_{L_1} \in \{a_1, a_2\}$ and $1_{L_2} \notin \{b_1, b_2\}$,
- (iii) $1_{L_1} \notin \{a_1, a_2\}$ and $1_{L_2} \in \{b_1, b_2\}$,
- (iv) $1_{L_1} \in \{a_1, a_2\}$ and $1_{L_2} \in \{b_1, b_2\}$.

- (i) $1_{L_1} \notin \{a_1, a_2\}$ and $1_{L_2} \notin \{b_1, b_2\}$: In this case, $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are minimal covers of L_1 and L_2 , respectively (see, Lemma 4.13 of [8]). Since $\text{ind}(L_1) = k_1$ and $\text{ind}(L_2) = k_2$, for every $i \in \{1, 2\}$, we have that $\text{ind}(\uparrow(a_i \vee a_i^*)) \leq k_1 - 1$ and $\text{ind}(\uparrow(b_i \vee b_i^*)) \leq k_2 - 1$. Therefore,

$$\begin{aligned} \text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) &= \max\{\text{ind}(\uparrow(a_i \vee a_i^*)), \text{ind}(\uparrow(b_i \vee b_i^*))\} \\ &\leq \max\{k_1 - 1, k_2 - 1\}. \end{aligned}$$

Thus, $\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\}$, for every $i \in \{1, 2\}$.

- (ii) $1_{L_1} \in \{a_1, a_2\}$ and $1_{L_2} \notin \{b_1, b_2\}$: Say $a_2 = 1_{L_1}$. (If $a_1 = 1_{L_1}$, then we work in a similar way.) In this case, $\{a_1, 1_{L_1}\}$ is not a minimal cover of L_1 and $\{b_1, b_2\}$ is a minimal cover of L_2 . Indeed, let $\{y_1, \dots, y_p\}$ be a cover of L_2 which refines $\{b_1, b_2\}$. We shall prove that $\{b_1, b_2\} \subseteq \{y_1, \dots, y_p\}$. Since $\{y_1, \dots, y_p\}$ refines $\{b_1, b_2\}$, for every $i \in \{1, \dots, p\}$, there exists $j \in \{1, 2\}$ such that $y_i \leq b_j$. We observe that there exists $i_0 \in \{1, \dots, p\}$ for which $y_{i_0} \leq b_2$. (If for every $i \in \{1, \dots, p\}$ we have $y_i \leq b_1$, then $\{y_1, \dots, y_p\}$ will not be a cover of L_2 .) We consider the set

$$W = \{(a_1, y_1), \dots, (a_1, y_{i_0-1}), (1_{L_1}, y_{i_0}), (a_1, y_{i_0+1}), \dots, (a_1, y_p)\}.$$

Then W is a cover of $L_1 \square L_2$ which refines the minimal cover $\{(a_1, b_1), (1_{L_1}, b_2)\}$. Therefore,

$$\{(a_1, b_1), (1_{L_1}, b_2)\} \subseteq \{(a_1, y_1), \dots, (a_1, y_{i_0-1}), (1_{L_1}, y_{i_0}), (a_1, y_{i_0+1}), \dots, (a_1, y_p)\}$$

and thus, $b_1 = y_i$, for some $i \in \{1, \dots, p\}$, $i \neq i_0$, and $b_2 = y_{i_0}$. Therefore, $\{b_1, b_2\} \subseteq \{y_1, \dots, y_p\}$. Since $\text{ind}(L_2) = k_2$, we have that $\text{ind}(\uparrow(b_i \vee b_i^*)) \leq k_2 - 1$, for every $i \in \{1, 2\}$ and thus,

$$\begin{aligned} \text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) &= \max\{\text{ind}(\uparrow(a_i \vee a_i^*)), \text{ind}(\uparrow(b_i \vee b_i^*))\} \\ &\leq \max\{\text{ind}(\uparrow(a_i \vee a_i^*)), k_2 - 1\}. \end{aligned}$$

By Lemma 14 since $\{1_{L_1}\}$ is a minimal cover of L_1 , we have that $\text{ind}(L_1) = 0$ and also since $\{1_{L_1}\}$ is a minimal cover of $\uparrow(a_1 \vee a_1^*)$, we have that $\text{ind}(\uparrow(a_1 \vee a_1^*)) = 0$. Moreover, we know that $\text{ind}(\uparrow(a_2 \vee a_2^*)) = -1 \leq k_1 - 1$. Thus, for every $i \in \{1, 2\}$ we have that

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1\}.$$

Therefore, $\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\}$, for every $i \in \{1, 2\}$.

(iii) $1_{L_1} \notin \{a_1, a_2\}$ and $1_{L_2} \in \{b_1, b_2\}$: This case is similar to (ii), considering Lemma 15.

(iv) $1_{L_1} \in \{a_1, a_2\}$ and $1_{L_2} \in \{b_1, b_2\}$: This case is similar to (ii), considering Lemma 16. $\square$

Proposition 18 Let L_1 and L_2 be two finite lattices. For any minimal cover $V = \{(a_1, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$, where $m > 2$, of $L_1 \square L_2$ for which $a_1 = 1_{L_1}$, $a_i \neq 1_{L_1}$ for every $i \in \{2, \dots, m\}$ and $\{b_1, \dots, b_m\}$ is a minimal cover of L_2 the following inequality holds

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{\text{ind}(L_1) - 1, \text{ind}(L_2) - 1, 0\},$$

for every $i \in \{1, 2, \dots, m\}$.

Proof We suppose that $\text{ind}(L_1) = k_1 \geq 0$ and $\text{ind}(L_2) = k_2 \geq 0$. We shall prove that

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\},$$

for every $i \in \{1, 2, \dots, m\}$. Since $\{b_1, \dots, b_m\}$ is a minimal cover of L_2 , $\text{ind}(\uparrow(b_i \vee b_i^*)) \leq k_2 - 1$, for every $i \in \{1, \dots, m\}$.

Also, by Lemma 14, $\{1_{L_1}\}$ is a minimal cover of L_1 and then, every cover of L_1 contains 1_{L_1} and $\text{ind}(L_1) = 0$. We know that $\text{ind}(\uparrow(a_1 \vee a_1^*)) = -1 \leq k_1 - 1$. We shall prove that $\text{ind}(\uparrow(a_i \vee a_i^*)) = 0$, for every $i \in \{2, \dots, m\}$. Indeed, let $i \in \{2, \dots, m\}$. Since $\{1_{L_1}\}$ is a minimal cover of $\uparrow(a_i \vee a_i^*)$, we have that $\text{ind}(\uparrow(a_i \vee a_i^*)) = 0$. Thus, for every $i \in \{1, 2, \dots, m\}$, we have

$$\begin{aligned} \text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) &= \max\{\text{ind}(\uparrow(a_i \vee a_i^*)), \text{ind}(\uparrow(b_i \vee b_i^*))\} \\ &\leq \max\{k_1 - 1, k_2 - 1\}. \end{aligned}$$

Altogether, $\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\}$, for every $i \in \{1, 2, \dots, m\}$.

$\square$

Proposition 19 Let L_1 and L_2 be two finite lattices. For any minimal cover $V = \{(a_1, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$, where $m > 2$, of $L_1 \square L_2$ for which $b_1 = 1_{L_2}$, $b_i \neq 1_{L_2}$, for every $i \in \{2, \dots, m\}$ and $\{a_1, \dots, a_m\}$ is a minimal cover of L_1 , the following inequality holds

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{\text{ind}(L_1) - 1, \text{ind}(L_2) - 1, 0\},$$

for every $i \in \{1, 2, \dots, m\}$.

Proof It is similar to the proof of Proposition 18. $\square$

Proposition 20 Let L_1 and L_2 be two finite lattices. For any minimal cover $V = \{(a_1, b_1), (a_2, b_2), \dots, (a_m, b_m)\}$, where $m > 2$, for which $1_{L_1} \notin \{a_1, \dots, a_m\}$ and $1_{L_2} \notin \{b_1, \dots, b_m\}$ such that $\{a_1, \dots, a_m\}$ and $\{b_1, \dots, b_m\}$ are minimal covers of L_1 and L_2 , respectively, the following inequality holds

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{\text{ind}(L_1) - 1, \text{ind}(L_2) - 1, 0\},$$

for every $i \in \{1, 2, \dots, m\}$.

Proof We suppose that $\text{ind}(L_1) = k_1 \geq 0$ and $\text{ind}(L_2) = k_2 \geq 0$. We shall prove that

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\},$$

for every $i \in \{1, 2, \dots, m\}$.

Since $\text{ind}(L_1) = k_1$ and $\text{ind}(L_2) = k_2$, we have that $\text{ind}(\uparrow(a_i \vee a_i^*)) \leq k_1 - 1$ and $\text{ind}(\uparrow(b_i \vee b_i^*)) \leq k_2 - 1$, for every $i \in \{1, \dots, m\}$. Thus, for every $i \in \{1, \dots, m\}$, we have that

$$\text{ind}(\uparrow((a_i, b_i) \vee (a_i, b_i)^*)) \leq \max\{k_1 - 1, k_2 - 1, 0\}.$$

Theorem 21 Let L_1 and L_2 be two finite lattices. If any minimal cover V of $L_1 \square L_2$ satisfies one of the conditions of Propositions 17, 18, 19 and 20, then the following inequality holds

$$\text{ind}(L_1 \square L_2) \leq \max\{\text{ind}(L_1), \text{ind}(L_2), 1\}.$$

Proof It follows by Theorem 2 and Propositions 17, 18, 19 and 20. $\square$

Remark 5 We observe that the small inductive dimension of the Cartesian product $L_1 \times L_2$ of two finite lattices L_1 and L_2 and of their rectangular product are different. In particular, both relations

$$\text{ind}(L_1 \times L_2) \leq \text{ind}(L_1 \square L_2) \text{ and } \text{ind}(L_1 \square L_2) \leq \text{ind}(L_1 \times L_2)$$

does not hold for all finite lattices L_1 and L_2 .

We consider the finite lattices, which are represented in Remark 4 (1). Then we have that $\text{ind}(L_1 \square L_2) = 1$ and since $\text{ind}(L_1) = \text{ind}(L_2) = 0$, by Theorem 8 we have that $\text{ind}(L_1 \times L_2) = 0$.

Moreover, for the lattices of Remark 4 (2) we have that $\text{ind}(L_2 \square L_1) = 0$ and $\text{ind}(L_2 \times L_1) = 1$.

4 Relationship to other Dimensions

In this section, we present some additional remarks for the small inductive dimension ind for the class of finite lattices.

We remind that the *height* of a finite lattice L is defined as follows:

$$h(L) = \max\{k : \text{there exist } a_0, \dots, a_k \in L \text{ such that } 0_L = a_0 < \dots < a_k = 1_L\}.$$

Example 2 (1) For the lattice L_1 of Figure 1 we have that $\text{ind}(L_1) = 0$ and $h(L_2) = 2$.

(2) For the totally ordered set $L = \{x_1, \dots, x_n, 0_L, 1_L\}$, we have that $h(L) = n + 1$ and $\text{ind}(L) = 0$.

Proposition 22 For every finite lattice L , $\text{ind}(L) < h(L)$.

Proof We prove the inequality by induction on $h(L)$. Clearly if $h(L) = 0$, then $L = \{1_L\}$ and thus, $\text{ind}(L) = -1$, proving that the relation of the proposition holds.

We suppose that the inequality holds for all finite lattices M with $h(M) < k$, $k \geq 1$, and we shall prove it for k . Let L be a finite lattice with $h(L) = k$, V be a minimal cover of L and $v \in V$. By Theorem 2 it suffices to prove that $\text{ind}(\uparrow(v \vee v^*)) \leq k - 2$. Since $\uparrow(v \vee v^*) \neq L$, we have that $h(\uparrow(v \vee v^*)) < h(L)$ and thus, $h(\uparrow(v \vee v^*)) \leq k - 1$. By inductive hypothesis we have that $\text{ind}(\uparrow(v \vee v^*)) < k - 1$, proving that $\text{ind}(L) \leq k - 1$ or equivalently $\text{ind}(L) < k$.

□

A non-empty subset F of a lattice $(L, \leq)$ is called *filter* if F has the following properties:

- (i) $F \neq L$,
- (ii) If $x \in F$ and $x \leq y$, then $y \in F$,
- (iii) If $x, y \in F$, then $x \wedge y \in F$.

A filter F is called *prime* if for every $x, y \in L$ with $x \vee y \in F$, we have $x \in F$ or $y \in F$. The set of all prime filters of a lattice L is usually denoted by $\mathcal{PF}(L)$.

If $\mathcal{PF}(L) \neq \emptyset$, then the *Krull dimension* (see [26]) of L is defined as follows:

$$\text{Kdim}(L) = \sup\{k : \text{there exist prime filters } F_0 \subset F_1 \subset \cdots \subset F_k\}.$$

Also, in the study [19], the Krull dimension is studied through join prime elements and matrices. An element $x \in L$ is said to be *join prime* if it is nonzero and the inequality $x \leq a \vee b$ implies $x \leq a$ or $x \leq b$, for all $a, b \in L$.

Example 3 We show by examples that the dimensions ind and Kdim for finite lattices are in general different.

(1) The small inductive dimension of both the diamond and the pentagon (see Figure 14) are equal to 0, but since they are not Boolean Algebras, their Krull dimensions are both not equal to 0 (see [26]).

(2) We consider the finite lattice $(L, \leq)$, represented by the diagram of Figure 15.

We have that $\text{ind}(L) = 2$ and $\text{Kdim}(L) = 1$. (The join prime elements of L are the elements x_1 and x_2 and thus, by Proposition 4.5 (5) of [19] we have that $\text{Kdim}(L) = 1$.)

Let L be a finite lattice. The *order* of a subset C of L , denoted by $\text{ord}(C)$, is defined to be k , where $k \in \mathbb{N}$ if and only if the infimum of any $k + 2$ distinct elements of C is 0_L and there exist $k + 1$ distinct elements of C whose infimum is not 0_L .

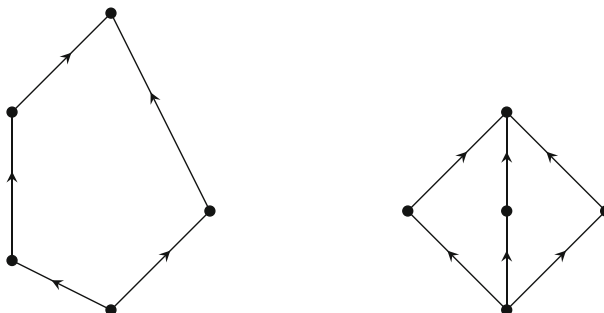
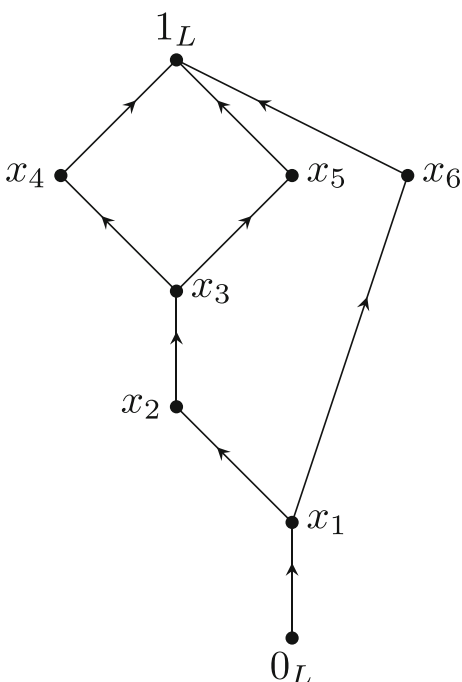


Fig. 14 The pentagon and the diamond

Fig. 15 The finite lattice $(L, \leq)$ 

The function $\dim$, called *covering dimension*, with domain the class of all finite lattices and range the set $\mathbb{N}$ is defined as follows:

1. $\dim(L) \leq k$, where $k \in \mathbb{N}$, if and only if for every cover C of L , there exists a cover R of L , refinement of C with $\text{ord}(R) \leq k$.
2. $\dim(L) = k$, where $k \in \mathbb{N}$, if $\dim(L) \leq k$ and $\dim(L) \not\leq k - 1$.

Remark 6 We show by examples that the dimensions ind and $\dim$ for finite lattices are in general different.

(1) We consider the lattice of Example 3 (2). Then $\text{ind}(L) = 2$ and $\dim(L) = 1$. (For the minimal covers $V_1 = \{x_2, x_6\}$ and $V_2 = \{x_4, x_5\}$ we have that $\text{ord}(V_1) = \text{ord}(V_2) = 1$.)

(2) We consider the lattice L represented by the diagram of Figure 16.

We have that the minimal cover of L is the set $\{x_2, x_3, x_4\}$, $\dim(L) = 2$ and $\text{ind}(L) = 1$.

Proposition 23 For every finite lattice L , if $\dim(L) = 0$, then $\text{ind}(L) = 0$.

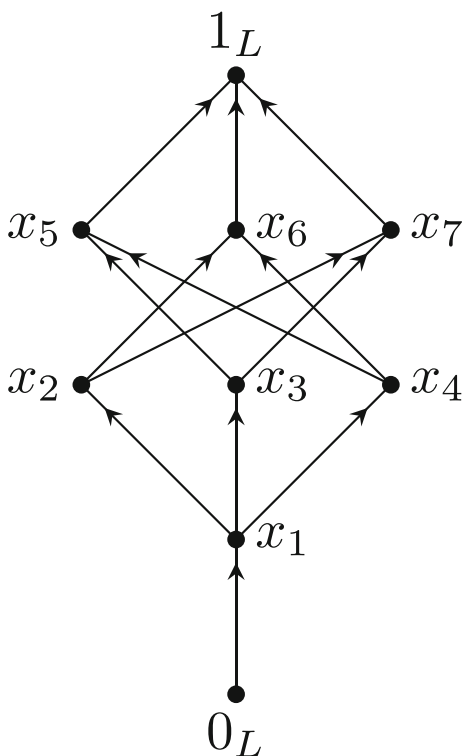
Proof Let L be a finite lattice, V be a minimal cover of L and $v \in V$. By Theorem 2 it suffices to prove that $\text{ind}(\uparrow(v \vee v^*)) = -1$. Since $\dim(L) = 0$, by [8] we have that $\text{ord}(V) = 0$. Therefore, $w \wedge v = 0_L$, for every $w \in V \setminus \{v\}$. Hence,

$$v^* = \bigvee \{y \in L : y \wedge v = 0_L\} \geq \bigvee \{w \in V : w \neq v\}$$

and

$$v^* \vee v \geq \bigvee \{w \in V : w \neq v\} \vee v = \bigvee V = 1_L,$$

proving that $\text{ind}(\uparrow(v \vee v^*)) = -1$. Therefore, $\text{ind}(L) = 0$. $\square$

Fig. 16 The finite lattice $(L, \leq)$ 

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Declarations

Not applicable

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